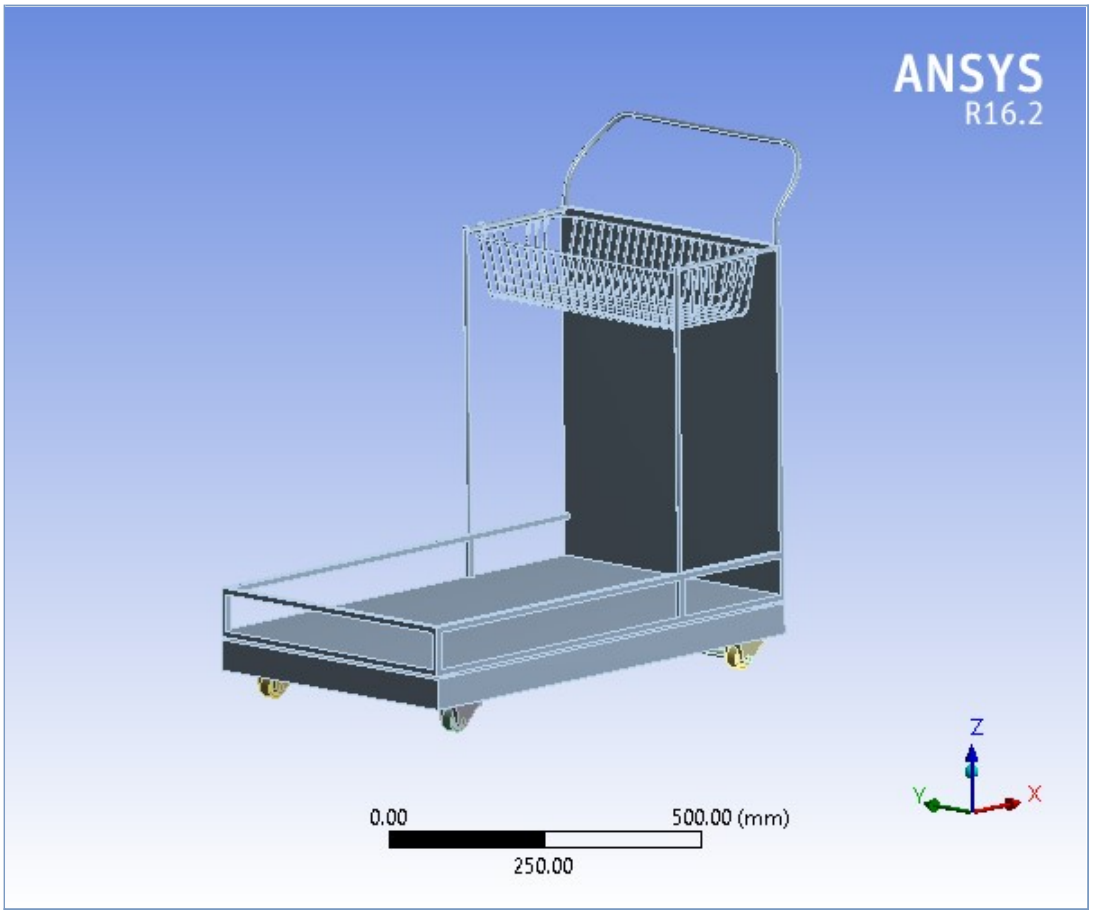
Project

First Saved	Thursday, July 3, 2025
Last Saved	Friday, July 4, 2025
Product Version	16.2 Release
Save Project Before Solution	No
Save Project After Solution	No



Contents

- **Units**
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Units

TABLE 1

Unit System	Metric (mm, t, N, s, mV, mA) Degrees rad/s Celsius
Angle	Degrees
Rotational Velocity	rad/s
Temperature	Celsius

Model (A4, C4)

Geometry

TABLE 2
Model (A4, C4) > Geometry

Object Name	*Geometry*
State	Fully Defined
Definition	
Source	C:\Users\Drz\Desktop\anex ansys\trolly anex ansys anal_files\dp0\SYS\DM\SYS.agdb
Type	DesignModeler
Length Unit	Meters
Element Control	Program Controlled
Display Style	Body Color
Bounding Box	
Length X	890.69 mm

Length Y	487.5 mm
Length Z	835.85 mm
Properties	
Volume	1.7897e+007 mm³
Mass	0.14049 t
Scale Factor Value	1.
Statistics	
Bodies	17
Active Bodies	17
Nodes	71220
Elements	32996
Mesh Metric	None
Basic Geometry Options	
Parameters	Yes
Parameter Key	DS
Attributes	No
Named Selections	No
Material Properties	No
Advanced Geometry Options	
Use Associativity	Yes
Coordinate Systems	No
Reader Mode Saves Updated File	No
Use Instances	Yes
Smart CAD Update	No
Compare Parts On Update	No
Attach File Via Temp File	Yes
Temporary Directory	C:\Users\Drz\AppData\Local\Temp
Analysis Type	3-D
Decompose Disjoint Geometry	Yes
Enclosure and Symmetry Processing	Yes

TABLE 3
Model (A4, C4) > Geometry > Parts

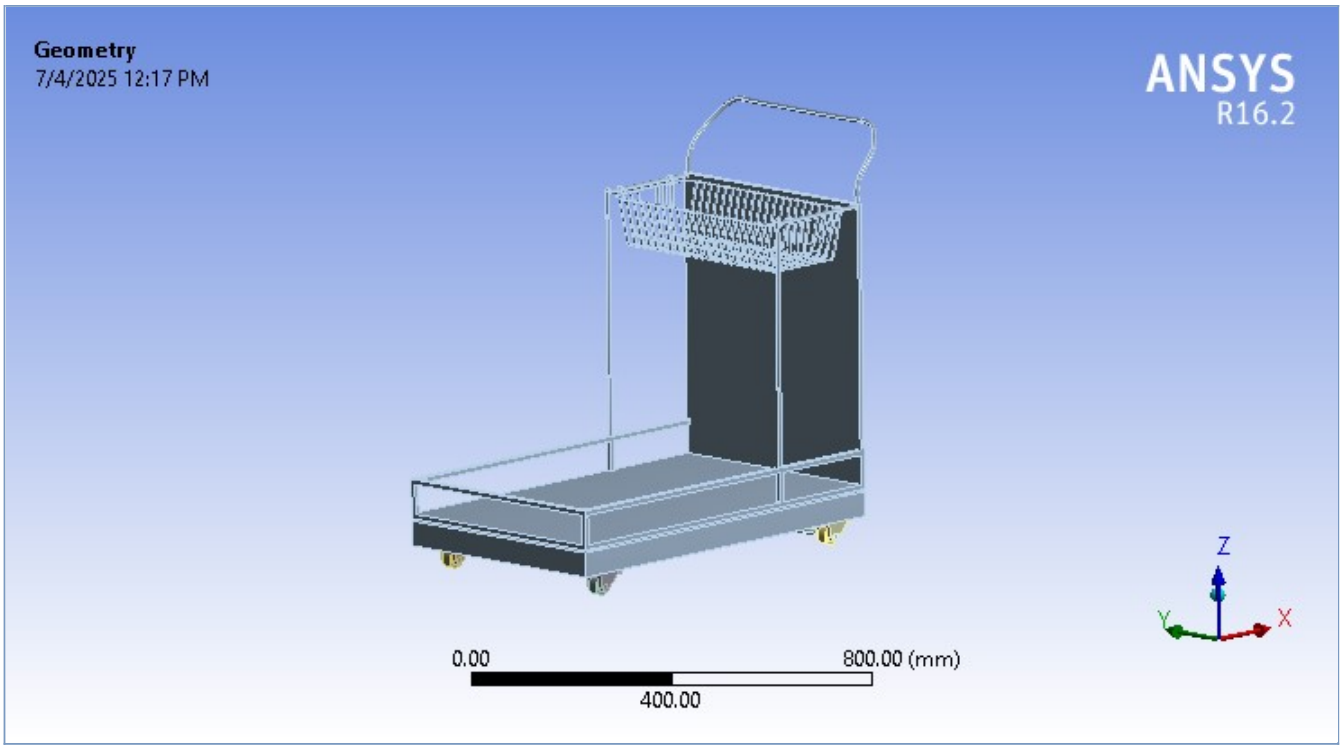
Object Name	*basket*	*mort*	*bat*	*back wheel*	*t*	*t*	*wheel*	*wheel*	*wheel shaft*	*shaft*	*shaft*
State	Meshed										
Graphics Properties											
Visible	Yes										
Transparency	1										
Definition											
Suppressed	No										
Stiffness Behavior	Flexible										
Coordinate System	Default Coordinate System										
Reference Temperature	By Environment										
Material											
Assignment	Structural Steel										
Nonlinear Effects	Yes										
Thermal Strain Effects	Yes										
Bounding Box											
Length X	225. mm	100. mm	180. mm	110. mm	36. mm		60. mm		100. mm	11. mm	
Length Y	487.5 mm	80. mm	140. mm	450. mm	20. mm		40. mm		445. mm	40. mm	
Length Z	117.25 mm	40. mm	30. mm	70. mm	36. mm		44.602 mm		45. mm	11. mm	
Properties											
Volume	2.0308e+005	3.2e+005	7.56e+005	1.6342e+005	17241 mm³		45400 mm³		78838	3801.3 mm³	

	mm³	mm³	mm³	mm³					mm³			
Mass	1.5942e-003 t	2.512e-003 t	5.9346e-003 t	1.2828e-003 t	1.3534e-004 t		3.5639e-004 t		6.1888e-004 t	2.984e-005 t		
Centroid X	-67. mm	-148.74 mm	-332.4 mm	-10.977 mm	-8.7413 mm		2.9021 mm		-624.8 mm	-663.74 mm		
Centroid Y	-213.94 mm			-216.44 mm	-416.44 mm	-11.441 mm	-416.44 mm	-11.441 mm	-213.94 mm	-416.44 mm	-11.441 mm	
Centroid Z	340.47 mm	-213. mm	-208. mm	-234.07 mm	-255.6 mm		-234.77 mm		-222.2 mm	-255.6 mm		
Moment of Inertia Ip1	35.088 t·mm²	1.6747 t·mm²	10.138 t·mm²	6.3563 t·mm²	1.6172e-002 t·mm²		9.6814e-002 t·mm²		2.3673 t·mm²	4.1819e-003 t·mm²		
Moment of Inertia Ip2	12.03 t·mm²	2.4283 t·mm²	16.469 t·mm²	0.83865 t·mm²	2.4276e-002 t·mm²		0.1273 t·mm²		0.18349 t·mm²	4.4677e-004 t·mm²		
Moment of Inertia Ip3	42.38 t·mm²	3.4331 t·mm²	25.717 t·mm²	6.0574 t·mm²	1.6172e-002 t·mm²		0.15631 t·mm²		2.3871 t·mm²	4.1819e-003 t·mm²		
Statistics												
Nodes	40013	578	571	1122	1338		1050	1048	1201	506		
Elements	18599	90	84	513	708	707	518	515	544	88		
Mesh Metric	None											

TABLE 4
Model (A4, C4) > Geometry > Parts

Object Name	*t*	*t*	*wheel*	*wheel*	*box*	*cari*
State	Meshed					
Graphics Properties						
Visible	Yes					
Transparency	1					
Definition						
Suppressed	No					
Stiffness Behavior	Flexible					
Coordinate System	Default Coordinate System					
Reference Temperature	By Environment					
Material						
Assignment	Structural Steel					
Nonlinear Effects	Yes					
Thermal Strain Effects	Yes					
Bounding Box						
Length X	36. mm		60. mm		100. mm	890.69 mm
Length Y	20. mm		40. mm		50. mm	480. mm
Length Z	36. mm		44.602 mm		55. mm	795.25 mm
Properties						
Volume	17241 mm³		45400 mm³		2.75e+005 mm³	1.5843e+007 mm³
Mass	1.3534e-004 t		3.5639e-004 t		2.1587e-003 t	0.12437 t
Centroid X	-663.74 mm		-652.1 mm		-491.5 mm	-269.96 mm
Centroid Y	-416.44 mm	-11.441 mm	-416.44 mm	-11.441 mm	-213.94 mm	-213.96 mm
Centroid Z	-255.6 mm		-234.77 mm		-220.5 mm	-119.34 mm
Moment of Inertia Ip1	1.6164e-002 t·mm²		9.6814e-002 t·mm²		0.99392 t·mm²	4485.5 t·mm²
Moment of Inertia Ip2	2.4263e-002 t·mm²		0.1273 t·mm²		2.3431 t·mm²	11283 t·mm²
Moment of Inertia Ip3	1.6164e-002 t·mm²		0.15631 t·mm²		2.2487 t·mm²	12123 t·mm²
Statistics						
Nodes	1326	1305	1061	1045	695	16517
Elements	700	685	525	513	112	8007
Mesh Metric	None					

FIGURE 1
Model (A4, C4) > Geometry > Image 2



Coordinate Systems

TABLE 5
Model (A4, C4) > Coordinate Systems > Coordinate System

Object Name	*Global Coordinate System*
State	Fully Defined
Definition	
Type	Cartesian
Coordinate System ID	0.
Origin	
Origin X	0. mm
Origin Y	0. mm
Origin Z	0. mm
Directional Vectors	
X Axis Data	[1. 0. 0.]
Y Axis Data	[0. 1. 0.]
Z Axis Data	[0. 0. 1.]

Connections

TABLE 6
Model (A4, C4) > Connections

Object Name	*Connections*
State	Fully Defined
Auto Detection	
Generate Automatic Connection On Refresh	Yes
Transparency	
Enabled	Yes

TABLE 7
Model (A4, C4) > Connections > Contacts

Object Name	*Contacts*
State	Fully Defined

Definition	
Connection Type	Contact
Scope	
Scoping Method	Geometry Selection
Geometry	All Bodies
Auto Detection	
Tolerance Type	Slider
Tolerance Slider	0.
Tolerance Value	3.2879 mm
Use Range	No
Face/Face	Yes
Face/Edge	No
Edge/Edge	No
Priority	Include All
Group By	Bodies
Search Across	Bodies
Statistics	
Connections	25
Active Connections	25

TABLE 8
Model (A4, C4) > Connections > Contacts > Contact Regions

Object Name	*Contact Region*	*Contact Region 2*	*Contact Region 3*	*Contact Region 4*	*Contact Region 5*	*Contact Region 6*	*Contact Region 7*	*Contact Region 8*	*Contact Region 9*	*Contact Region 10*	*Contact Region 11*
State	Fully Defined										
Scope											
Scoping Method	Geometry Selection										
Contact	17 Faces	1 Face			2 Faces				1 Face	7 Faces	
Target	4 Faces	1 Face			2 Faces		4 Faces		1 Face	3 Faces	
Contact Bodies	basket	mort		bat	back wheel					t	
Target Bodies	cari	back wheel	cari		t		wheel		cari	wheel	
Definition											
Type	Bonded										
Scope Mode	Automatic										
Behavior	Program Controlled										
Trim Contact	Program Controlled										
Trim Tolerance	3.2879 mm										
Suppressed	No										
Advanced											
Formulation	Program Controlled										
Detection Method	Program Controlled										
Penetration Tolerance	Program Controlled										
Elastic Slip Tolerance	Program Controlled										
Normal Stiffness	Program Controlled										
Update Stiffness	Program Controlled										
Pinball Region	Program Controlled										
Geometric Modification											
Contact											

Geometry Correction	None
Target Geometry Correction	None

TABLE 9
Model (A4, C4) > Connections > Contacts > Contact Regions

Object Name	*Contact Region 12*	*Contact Region 13*	*Contact Region 14*	*Contact Region 15*	*Contact Region 16*	*Contact Region 17*	*Contact Region 18*	*Contact Region 19*	*Contact Region 20*	*Contact Region 21*	*Contact Region 22*
State	Fully Defined										
Scope											
Scoping Method	Geometry Selection										
Contact	3 Faces		2 Faces		1 Face	2 Faces				7 Faces	
Target	3 Faces		2 Faces		1 Face	2 Faces	4 Faces	2 Faces	4 Faces	3 Faces	
Contact Bodies	wheel		wheel shaft			shaft				t	
Target Bodies	cari		wheel		box	t	wheel	t	wheel		
Definition											
Type	Bonded										
Scope Mode	Automatic										
Behavior	Program Controlled										
Trim Contact	Program Controlled										
Trim Tolerance	3.2879 mm										
Suppressed	No										
Advanced											
Formulation	Program Controlled										
Detection Method	Program Controlled										
Penetration Tolerance	Program Controlled										
Elastic Slip Tolerance	Program Controlled										
Normal Stiffness	Program Controlled										
Update Stiffness	Program Controlled										
Pinball Region	Program Controlled										
Geometric Modification											
Contact Geometry Correction	None										
Target Geometry Correction	None										

TABLE 10
Model (A4, C4) > Connections > Contacts > Contact Regions

Object Name	*Contact Region 23*	*Contact Region 24*	*Contact Region 25*
State	Fully Defined		
Scope			
Scoping Method	Geometry Selection		
Contact	3 Faces		1 Face
Target	3 Faces		1 Face
Contact Bodies	wheel		box
Target Bodies	cari		

Definition	
Type	Bonded
Scope Mode	Automatic
Behavior	Program Controlled
Trim Contact	Program Controlled
Trim Tolerance	3.2879 mm
Suppressed	No
Advanced	
Formulation	Program Controlled
Detection Method	Program Controlled
Penetration Tolerance	Program Controlled
Elastic Slip Tolerance	Program Controlled
Normal Stiffness	Program Controlled
Update Stiffness	Program Controlled
Pinball Region	Program Controlled
Geometric Modification	
Contact Geometry Correction	None
Target Geometry Correction	None

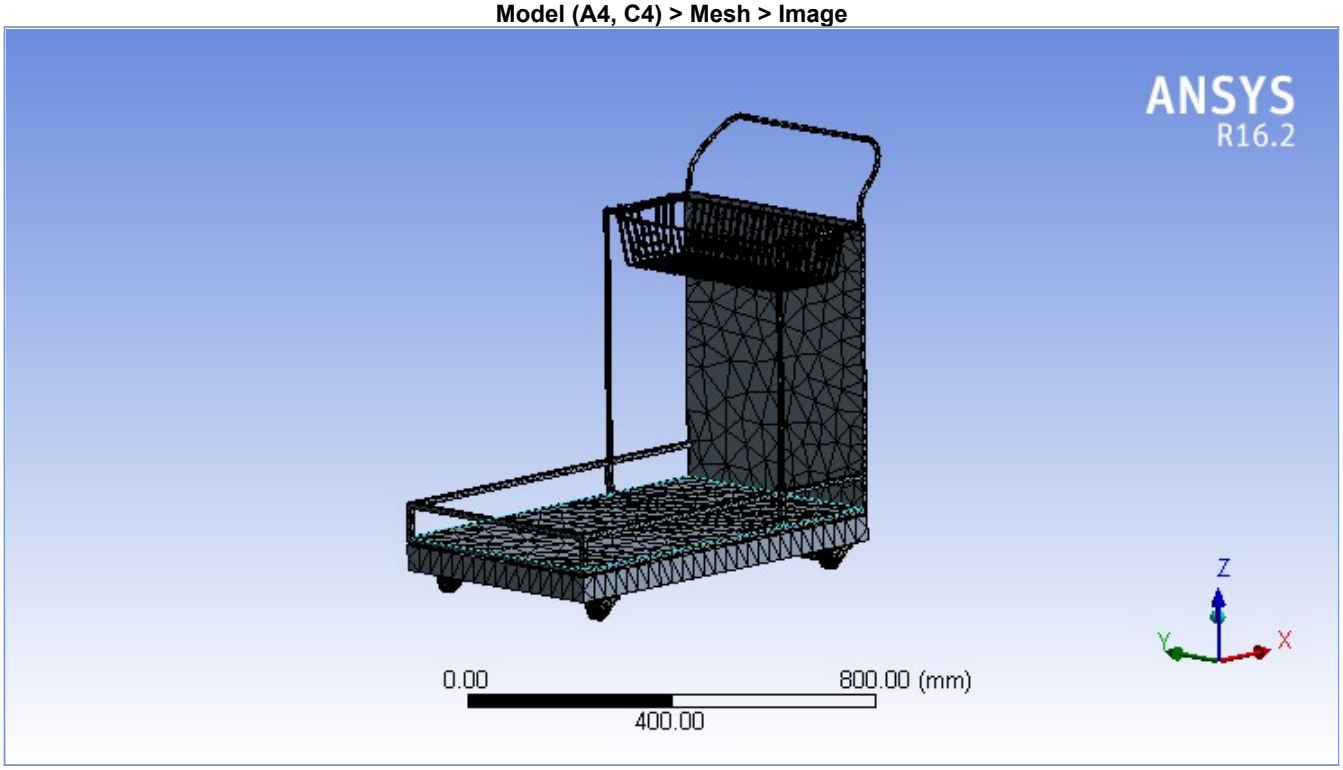
Mesh

TABLE 11
Model (A4, C4) > Mesh

Object Name	*Mesh*
State	Solved
Display	
Display Style	Body Color
Defaults	
Physics Preference	Mechanical
Relevance	0
Sizing	
Use Advanced Size Function	Off
Relevance Center	Coarse
Element Size	Default
Initial Size Seed	Active Assembly
Smoothing	Medium
Transition	Fast
Span Angle Center	Coarse
Minimum Edge Length	3.68e-002 mm
Inflation	
Use Automatic Inflation	None
Inflation Option	Smooth Transition
Transition Ratio	0.272
Maximum Layers	5
Growth Rate	1.2
Inflation Algorithm	Pre
View Advanced Options	No
Patch Conforming Options	
Triangle Surface Mesher	Program Controlled
Patch Independent Options	
Topology Checking	No
Advanced	
Number of CPUs for Parallel Part Meshing	Program Controlled
Shape Checking	Standard Mechanical
Element Midside Nodes	Program Controlled
Straight Sided Elements	No
Number of Retries	Default (4)
Extra Retries For Assembly	Yes

Rigid Body Behavior	Dimensionally Reduced
Mesh Morphing	Disabled
Defeaturing	
Pinch Tolerance	Please Define
Generate Pinch on Refresh	No
Automatic Mesh Based Defeaturing	On
Defeaturing Tolerance	Default
Statistics	
Nodes	71220
Elements	32996
Mesh Metric	None

FIGURE 2
Model (A4, C4) > Mesh > Image



Static Structural (A5)

TABLE 12
Model (A4, C4) > Analysis

Object Name	*Static Structural (A5)*
State	Solved
Definition	
Physics Type	Structural
Analysis Type	Static Structural
Solver Target	Mechanical APDL
Options	
Environment Temperature	22. °C
Generate Input Only	No

TABLE 13
Model (A4, C4) > Static Structural (A5) > Analysis Settings

Object Name	*Analysis Settings*
State	Fully Defined
Restart Analysis	

Restart Type	Program Controlled
Status	Done
Step Controls	
Number Of Steps	1.
Current Step Number	1.
Step End Time	1. s
Auto Time Stepping	Program Controlled
Solver Controls	
Solver Type	Program Controlled
Weak Springs	Program Controlled
Solver Pivot Checking	Program Controlled
Large Deflection	Off
Inertia Relief	Off
Restart Controls	
Generate Restart Points	Program Controlled
Retain Files After Full Solve	Yes
Nonlinear Controls	
Newton-Raphson Option	Program Controlled
Force Convergence	Program Controlled
Moment Convergence	Program Controlled
Displacement Convergence	Program Controlled
Rotation Convergence	Program Controlled
Line Search	Program Controlled
Stabilization	Off
Output Controls	
Stress	Yes
Strain	Yes
Nodal Forces	No
Contact Miscellaneous	No
General Miscellaneous	No
Store Results At	All Time Points
Analysis Data Management	
Solver Files Directory	C:\Users\Drz\Desktop\anex ansys\trolly anex ansys anal_files\dp0\SYS\MECH\
Future Analysis	Prestressed analysis
Scratch Solver Files Directory	
Save MAPDL db	No
Delete Unneeded Files	Yes
Nonlinear Solution	No
Solver Units	Active System
Solver Unit System	nmm

TABLE 14
Model (A4, C4) > Static Structural (A5) > Accelerations

Object Name	*Standard Earth Gravity*
State	Fully Defined
Scope	
Geometry	All Bodies
Definition	
Coordinate System	Global Coordinate System
X Component	0. mm/s² (ramped)
Y Component	0. mm/s² (ramped)
Z Component	-9806.6 mm/s² (ramped)
Suppressed	No
Direction	-Z Direction

FIGURE 3
Model (A4, C4) > Static Structural (A5) > Standard Earth Gravity

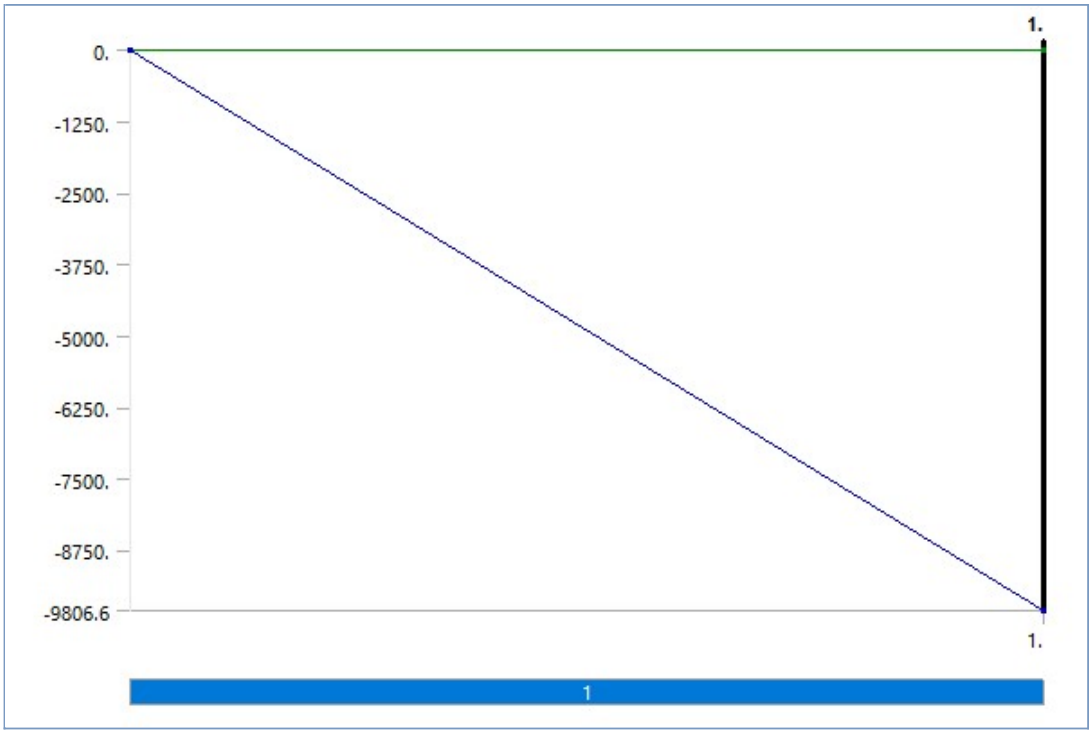


TABLE 15
Model (A4, C4) > Static Structural (A5) > Loads

Object Name	*Fixed Support*	*Force*	*Force 2*	*Fixed Support 2*
State	Fully Defined			
Scope				
Scoping Method	Geometry Selection			
Geometry	4 Faces	1 Face	27 Faces	24 Faces
Definition				
Type	Fixed Support	Force		Fixed Support
Suppressed	No			
Define By		Vector		
Magnitude		11000 N (ramped)	75. N (ramped)	
Direction		Defined		

FIGURE 4
Model (A4, C4) > Static Structural (A5) > Force

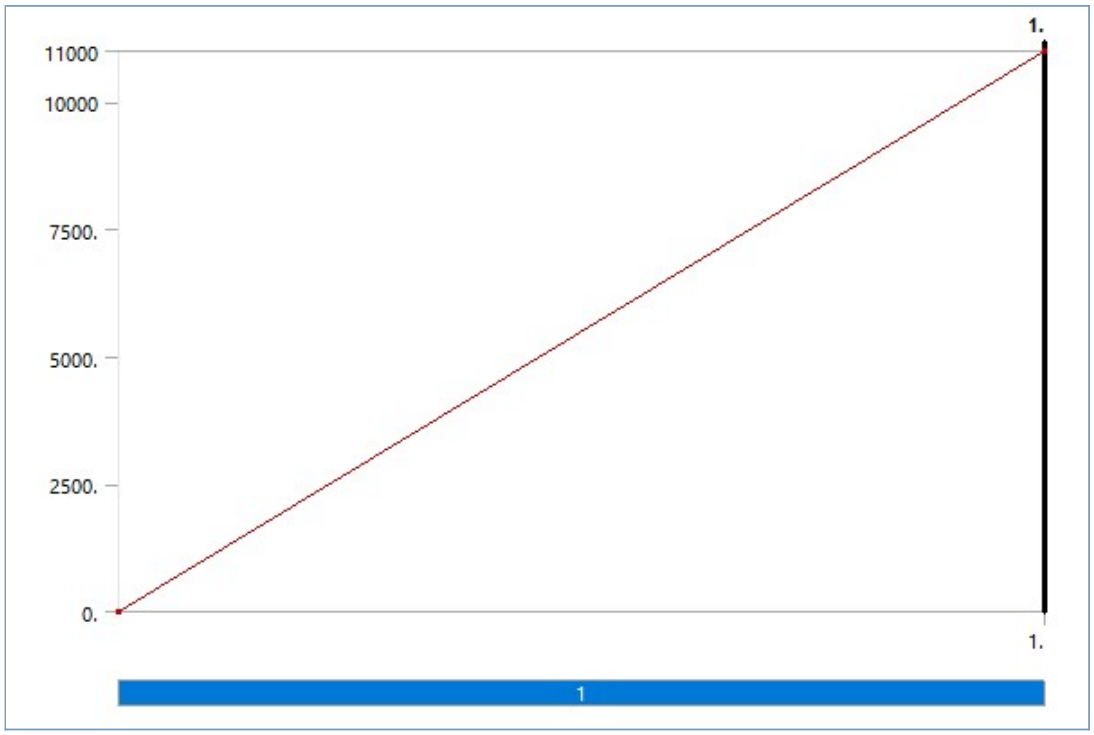


FIGURE 5
Model (A4, C4) > Static Structural (A5) > Force 2

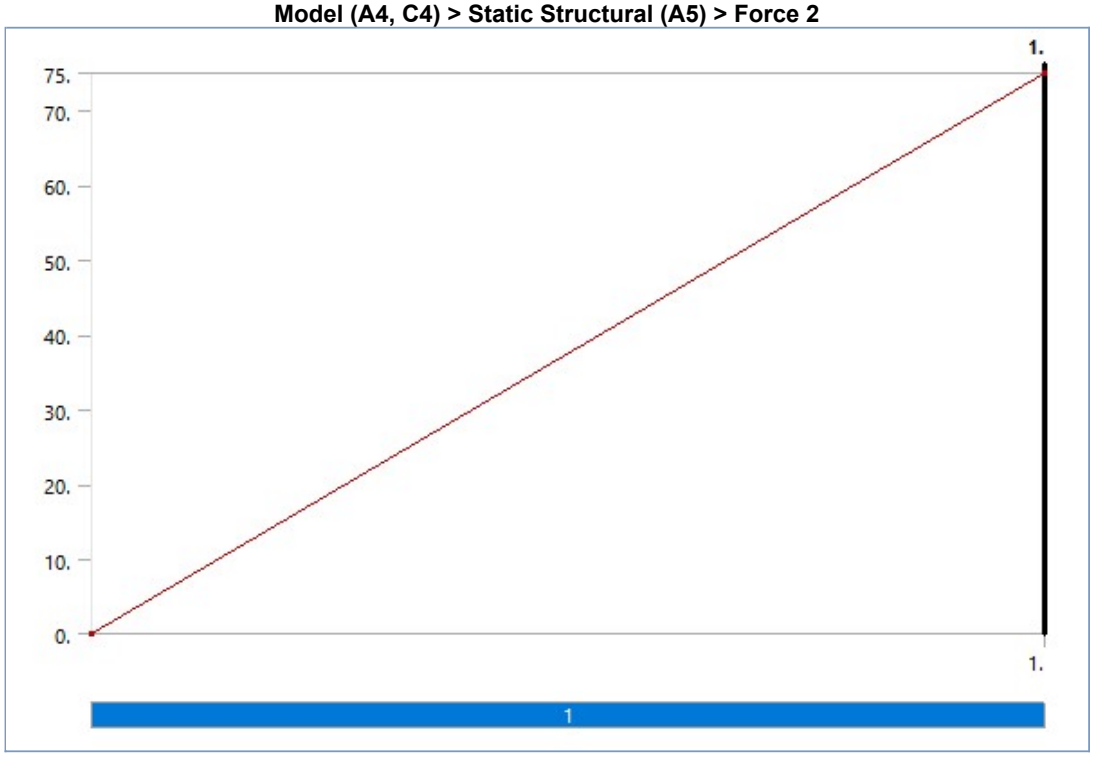
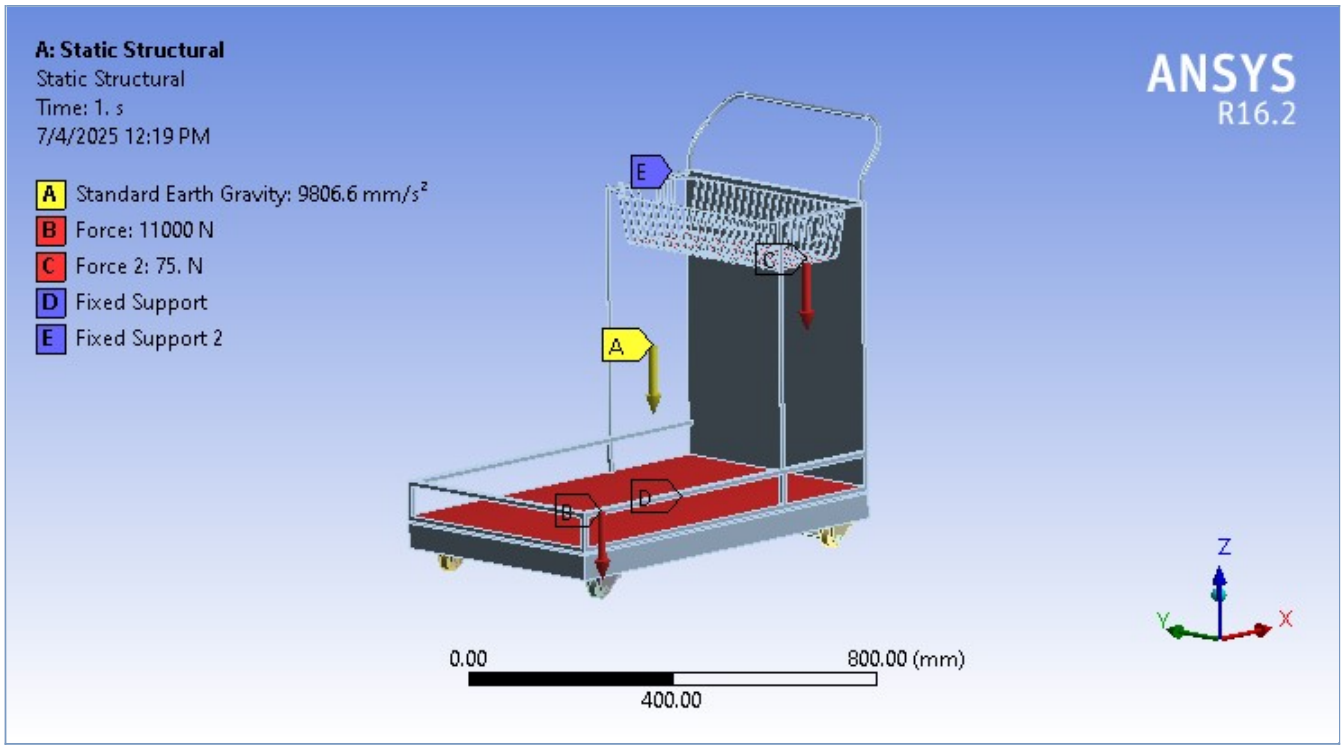


FIGURE 6
Model (A4, C4) > Static Structural (A5) > Image



Solution (A6)

TABLE 16
Model (A4, C4) > Static Structural (A5) > Solution

Object Name	*Solution (A6)*
State	Solved
Adaptive Mesh Refinement	
Max Refinement Loops	1.
Refinement Depth	2.
Information	
Status	Done
Post Processing	
Calculate Beam Section Results	No

TABLE 17
Model (A4, C4) > Static Structural (A5) > Solution (A6) > Solution Information

Object Name	*Solution Information*
State	Solved
Solution Information	
Solution Output	Solver Output
Newton-Raphson Residuals	0
Update Interval	2.5 s
Display Points	All
FE Connection Visibility	
Activate Visibility	Yes
Display	All FE Connectors
Draw Connections Attached To	All Nodes
Line Color	Connection Type
Visible on Results	No
Line Thickness	Single
Display Type	Lines

TABLE 18
Model (A4, C4) > Static Structural (A5) > Solution (A6) > Results

Object Name	*Total Deformation*	*Equivalent Stress*
State	Solved	
Scope		
Scoping Method	Geometry Selection	
Geometry	All Bodies	
Definition		
Type	Total Deformation	Equivalent (von-Mises) Stress
By	Time	
Display Time	Last	
Calculate Time History	Yes	
Identifier		
Suppressed	No	
Results		
Minimum	0. mm	0. MPa
Maximum	2.462 mm	128.58 MPa
Minimum Occurs On	basket	
Maximum Occurs On	basket	
Information		
Time	1. s	
Load Step	1	
Substep	1	
Iteration Number	1	
Integration Point Results		
Display Option		Averaged
Average Across Bodies		No

FIGURE 7
Model (A4, C4) > Static Structural (A5) > Solution (A6) > Total Deformation

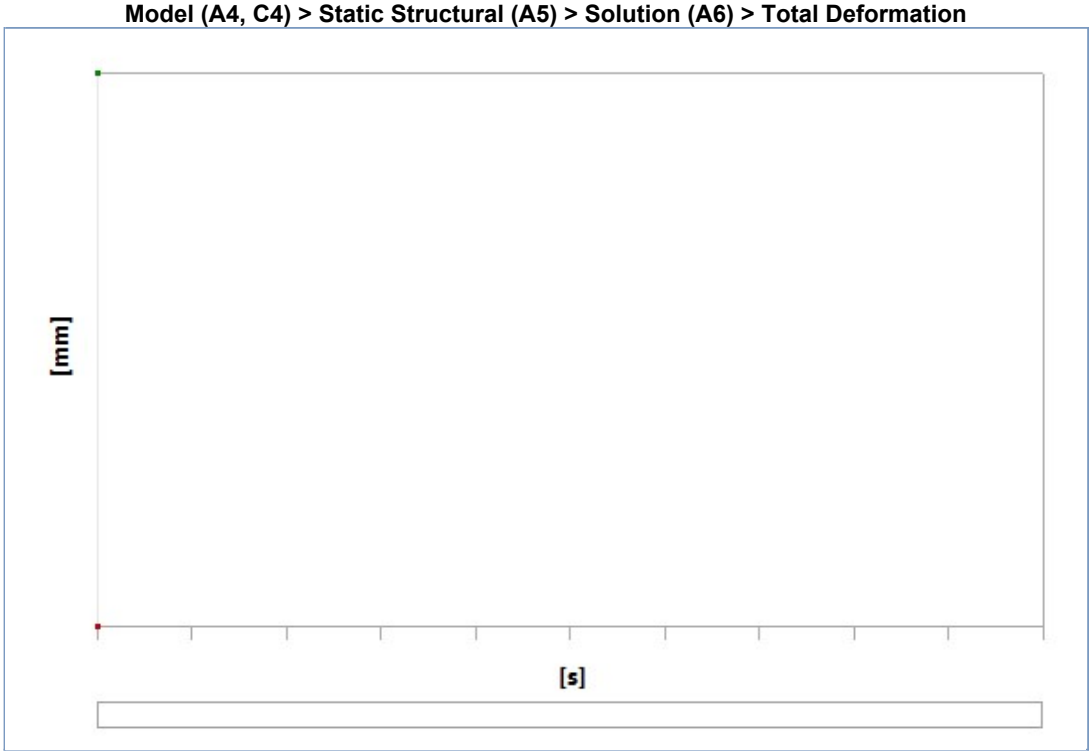


TABLE 19
Model (A4, C4) > Static Structural (A5) > Solution (A6) > Total Deformation

Time [s]	Minimum [mm]	Maximum [mm]
1.	0.	2.462

FIGURE 8

Model (A4, C4) > Static Structural (A5) > Solution (A6) > Total Deformation > Image

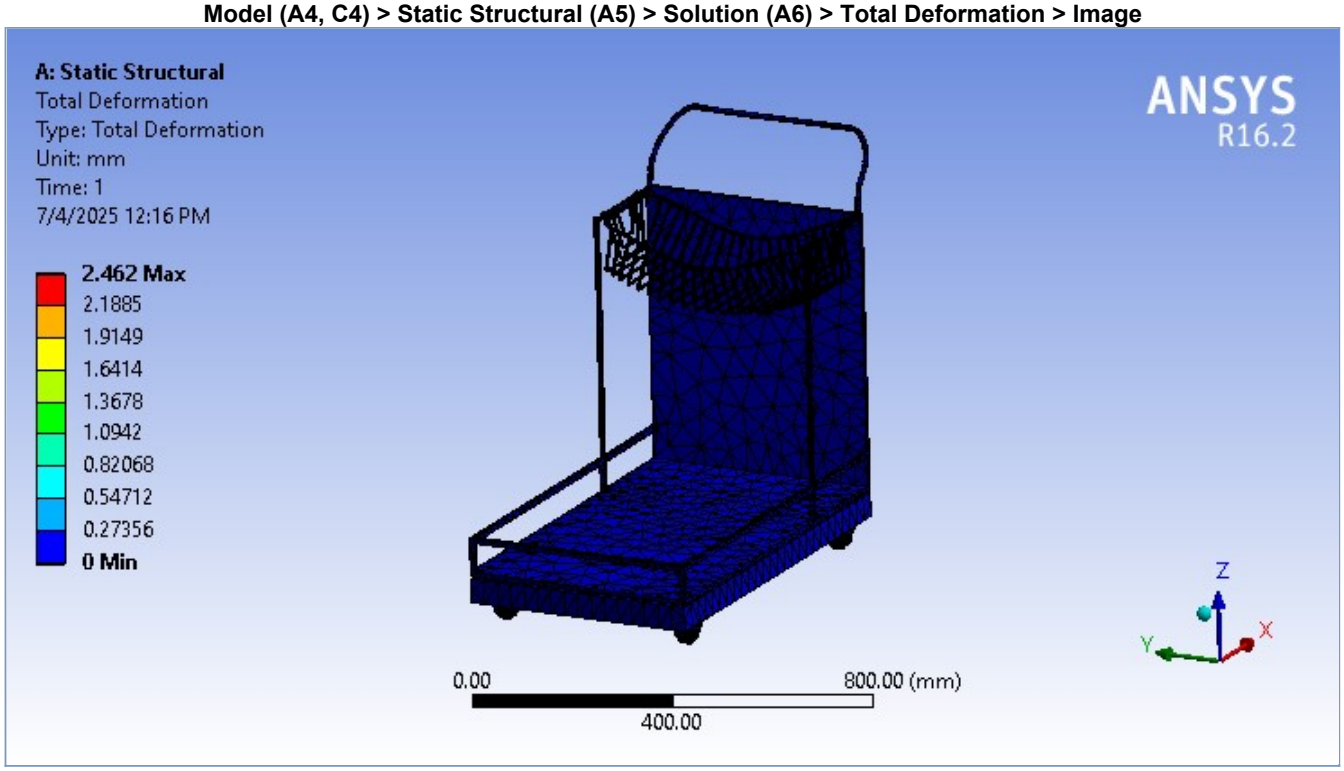


FIGURE 9
Model (A4, C4) > Static Structural (A5) > Solution (A6) > Equivalent Stress

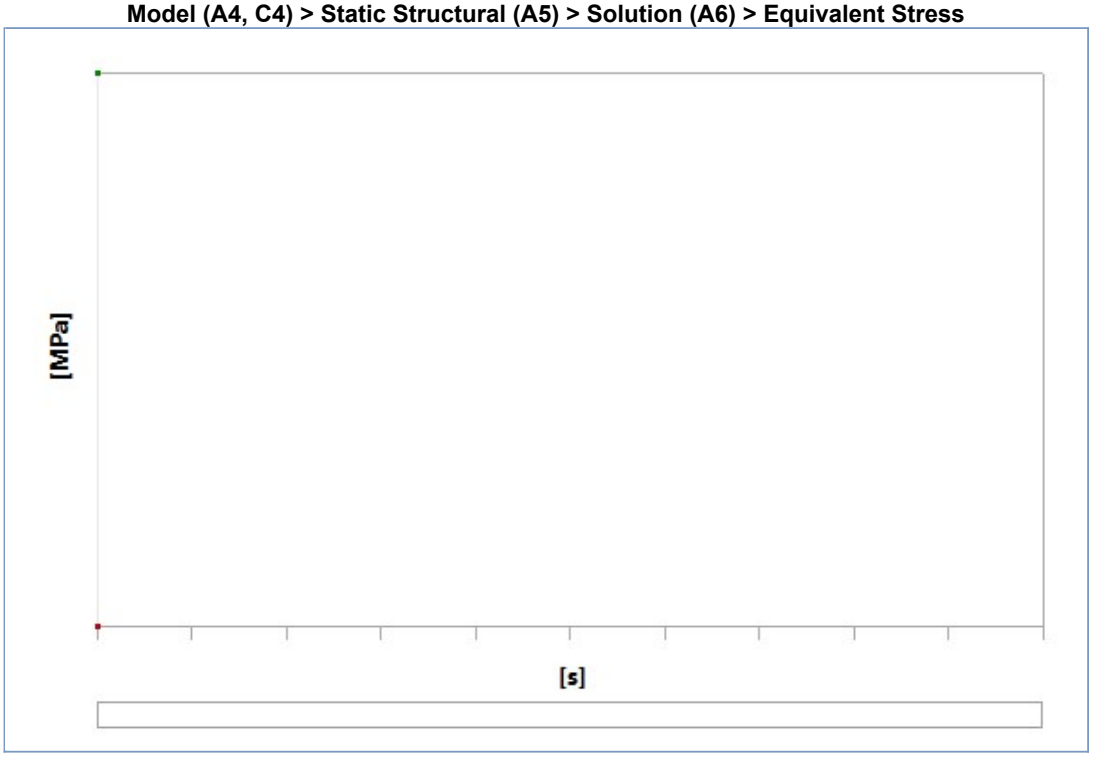


TABLE 20
Model (A4, C4) > Static Structural (A5) > Solution (A6) > Equivalent Stress

Time [s]	Minimum [MPa]	Maximum [MPa]
1.	0.	128.58

FIGURE 10
Model (A4, C4) > Static Structural (A5) > Solution (A6) > Equivalent Stress > Image

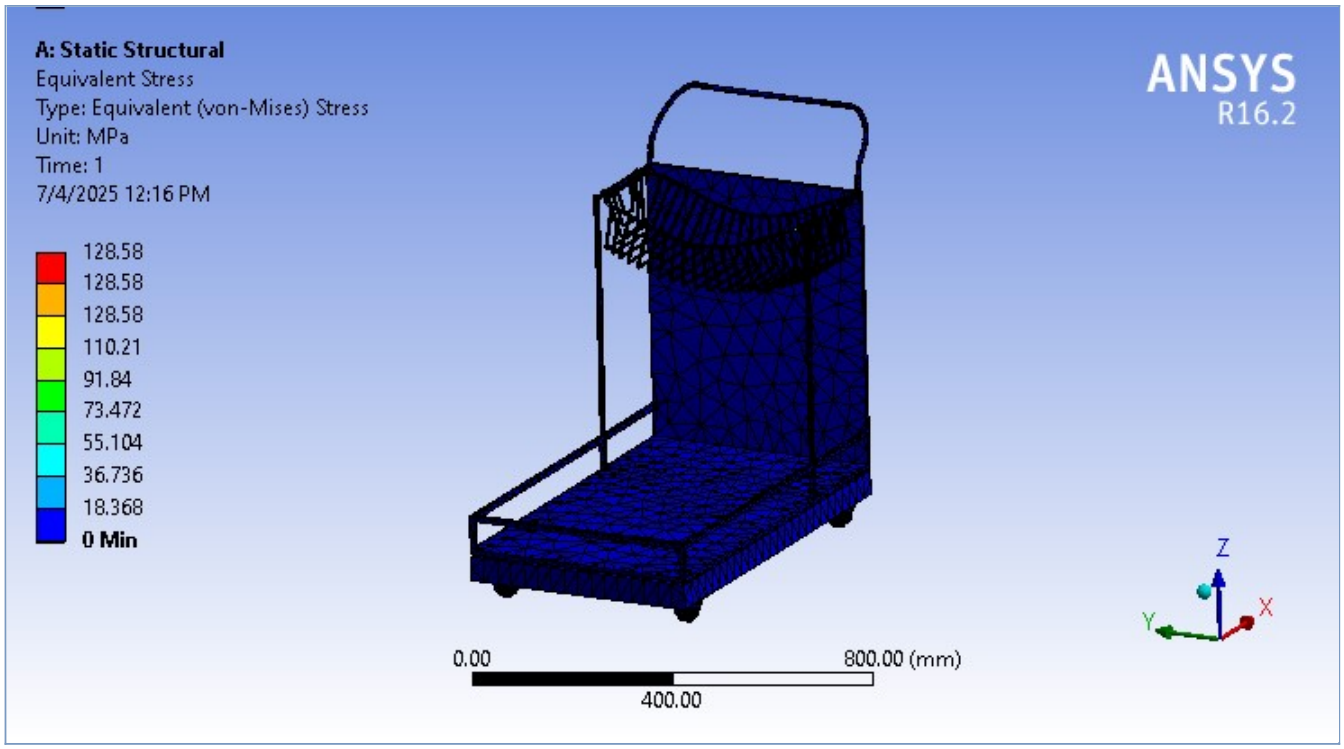


TABLE 21
Model (A4, C4) > Static Structural (A5) > Solution (A6) > Stress Safety Tools

Object Name	*Stress Tool*
State	Solved
Definition	
Theory	Max Equivalent Stress
Stress Limit Type	Tensile Yield Per Material

TABLE 22
Model (A4, C4) > Static Structural (A5) > Solution (A6) > Stress Tool > Results

Object Name	*Safety Factor*
State	Solved
Scope	
Scoping Method	Geometry Selection
Geometry	All Bodies
Definition	
Type	Safety Factor
By	Time
Display Time	Last
Calculate Time History	Yes
Identifier	
Suppressed	No
Integration Point Results	
Display Option	Averaged
Average Across Bodies	No
Results	
Minimum	1.9444
Minimum Occurs On	basket
Information	
Time	1. s
Load Step	1
Substep	1
Iteration Number	1

FIGURE 11
Model (A4, C4) > Static Structural (A5) > Solution (A6) > Stress Tool > Safety Factor

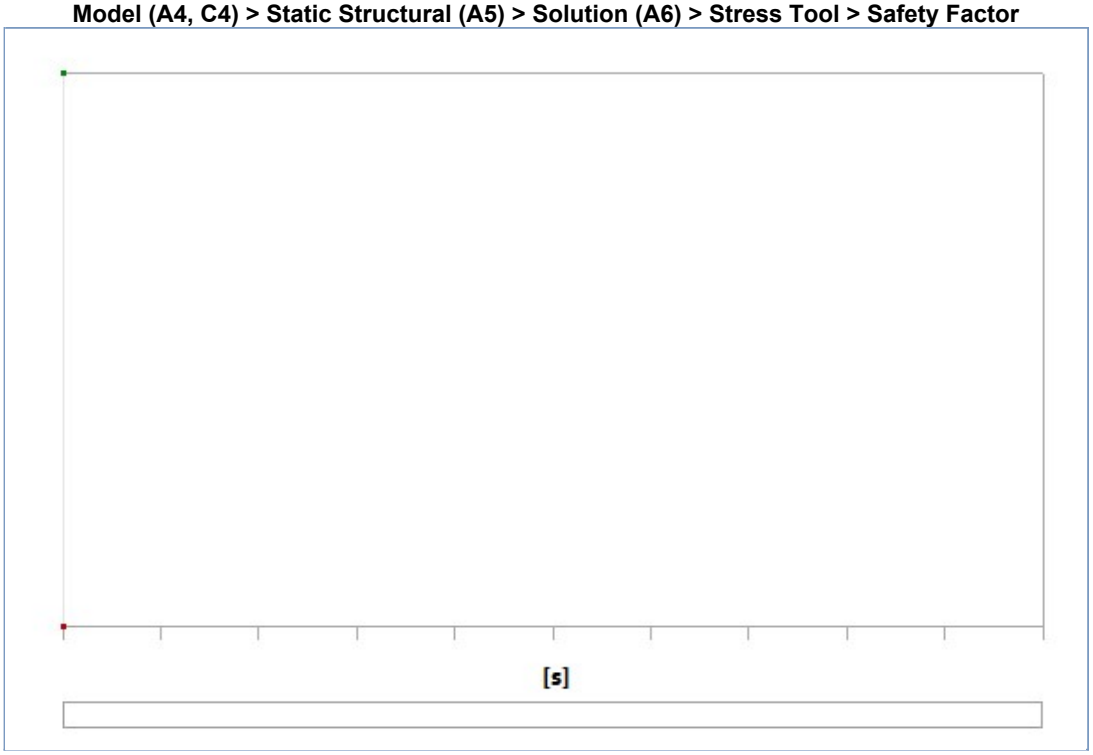
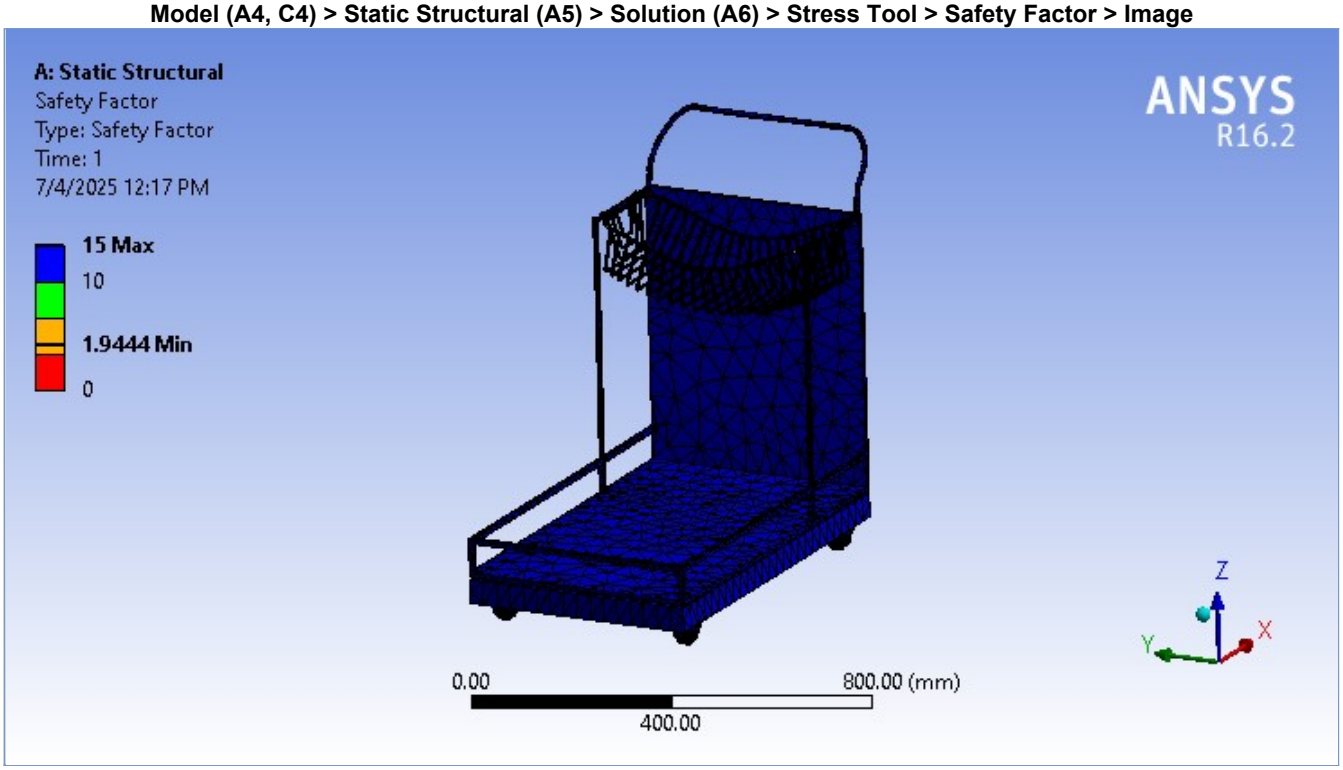


TABLE 23
Model (A4, C4) > Static Structural (A5) > Solution (A6) > Stress Tool > Safety Factor

Time [s]	Minimum	Maximum
1.	1.9444	15.

FIGURE 12
Model (A4, C4) > Static Structural (A5) > Solution (A6) > Stress Tool > Safety Factor > Image



Eigenvalue Buckling (C5)

TABLE 24
Model (A4, C4) > Analysis

Object Name	*Eigenvalue Buckling (C5)*
State	Solved
Definition	
Physics Type	Structural
Analysis Type	Eigenvalue Buckling
Solver Target	Mechanical APDL
Options	
Generate Input Only	No

TABLE 25
Model (A4, C4) > Eigenvalue Buckling (C5) > Initial Condition

Object Name	*Pre-Stress (Static Structural)*
State	Fully Defined
Definition	
Pre-Stress Environment	Nonlinear Static Structural
Pre-Stress Define By	Program Controlled
Reported Loadstep	Last
Reported Substep	Last
Reported Time	End Time
Contact Status	Use True Status

TABLE 26
Model (A4, C4) > Eigenvalue Buckling (C5) > Analysis Settings

Object Name	*Analysis Settings*
State	Fully Defined
Options	
Max Modes to Find	2.
Keep Pre-Stress Load-Pattern	Yes
Solver Controls	
Solver Type	Program Controlled
Include Negative Load Multiplier	Program Controlled
Output Controls	
Stress	No
Strain	No
General Miscellaneous	No
Analysis Data Management	
Solver Files Directory	C:\Users\Drz\Desktop\anex ansys\trolly anex ansys anal_files\dp0\SYS-2\MECH\
Future Analysis	None
Scratch Solver Files Directory	
Save MAPDL db	No
Delete Unneeded Files	Yes
Solver Units	Active System
Solver Unit System	nmm

Solution (C6)

TABLE 27
Model (A4, C4) > Eigenvalue Buckling (C5) > Solution

Object Name	*Solution (C6)*
State	Solved
Adaptive Mesh Refinement	
Max Refinement Loops	1.
Refinement Depth	2.
Information	

Status	Done
Post Processing	
Calculate Beam Section Results	No

FIGURE 13
Model (A4, C4) > Eigenvalue Buckling (C5) > Solution (C6)

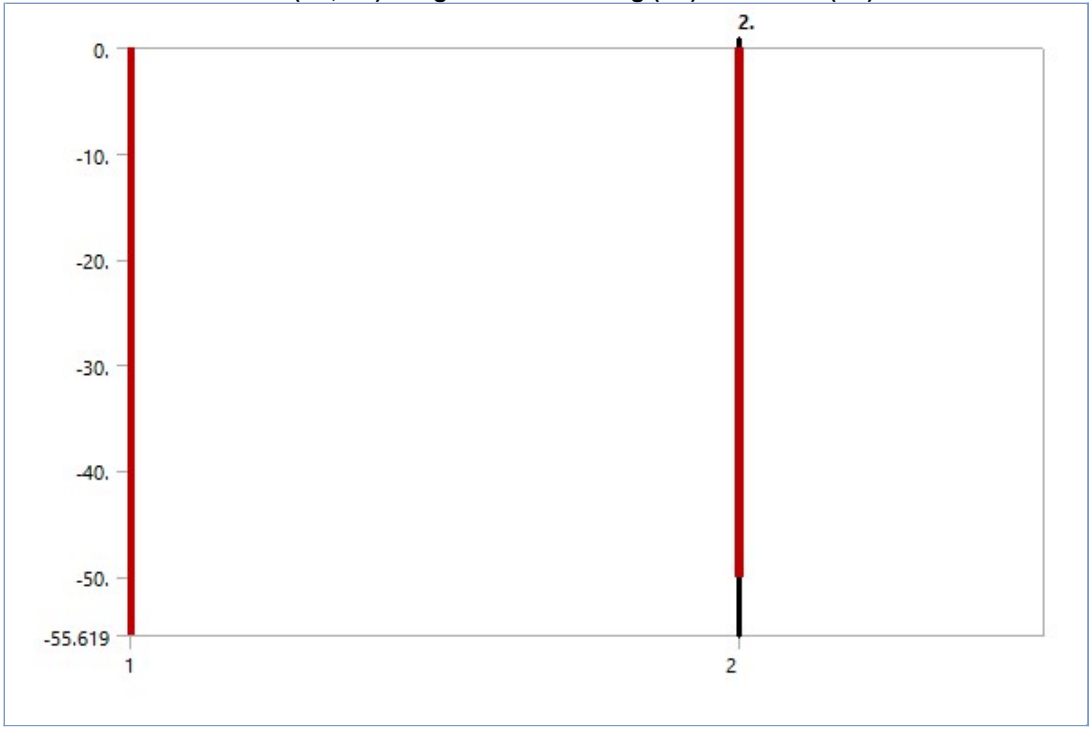


TABLE 28
Model (A4, C4) > Eigenvalue Buckling (C5) > Solution (C6)

Mode	Load Multiplier
1.	-55.619
2.	-50.234

TABLE 29
Model (A4, C4) > Eigenvalue Buckling (C5) > Solution (C6) > Solution Information

Object Name	*Solution Information*
State	Solved
Solution Information	
Solution Output	Solver Output
Newton-Raphson Residuals	0
Update Interval	2.5 s
Display Points	All
FE Connection Visibility	
Activate Visibility	Yes
Display	All FE Connectors
Draw Connections Attached To	All Nodes
Line Color	Connection Type
Visible on Results	No
Line Thickness	Single
Display Type	Lines

TABLE 30
Model (A4, C4) > Eigenvalue Buckling (C5) > Solution (C6) > Results

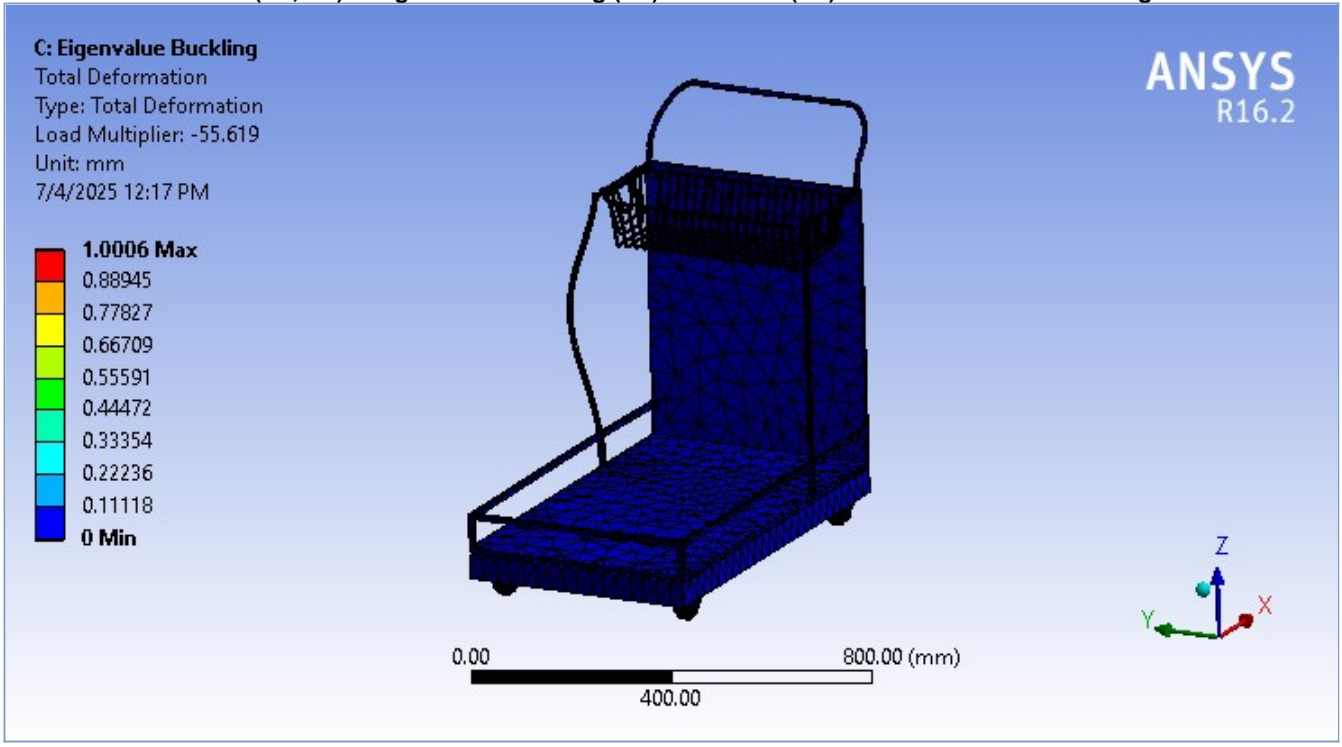
Object Name	*Total Deformation*
State	Solved

Scope	
Scoping Method	Geometry Selection
Geometry	All Bodies
Definition	
Type	Total Deformation
Mode	1.
Identifier	
Suppressed	No
Results	
Load Multiplier	-55.619
Minimum	0. mm
Maximum	1.0006 mm
Minimum Occurs On	basket
Maximum Occurs On	cari

TABLE 31
Model (A4, C4) > Eigenvalue Buckling (C5) > Solution (C6) > Total Deformation

Mode	Load Multiplier
1.	-55.619
2.	-50.234

FIGURE 14
Model (A4, C4) > Eigenvalue Buckling (C5) > Solution (C6) > Total Deformation > Image



Material Data

Structural Steel

TABLE 32
Structural Steel > Constants

Density	7.85e-009 tonne mm^-3
Coefficient of Thermal Expansion	1.2e-005 C^-1
Specific Heat	4.34e+008 mJ tonne^-1 C^-1
Thermal Conductivity	6.05e-002 W mm^-1 C^-1

Resistivity	1.7e-004 ohm mm

TABLE 33
Structural Steel > Compressive Ultimate Strength

Compressive Ultimate Strength MPa
0

TABLE 34
Structural Steel > Compressive Yield Strength

Compressive Yield Strength MPa
250

TABLE 35
Structural Steel > Tensile Yield Strength

Tensile Yield Strength MPa
250

TABLE 36
Structural Steel > Tensile Ultimate Strength

Tensile Ultimate Strength MPa
460

TABLE 37
Structural Steel > Isotropic Secant Coefficient of Thermal Expansion

Reference Temperature C
22

TABLE 38
Structural Steel > Alternating Stress Mean Stress

Alternating Stress MPa	Cycles	Mean Stress MPa
3999	10	0
2827	20	0
1896	50	0
1413	100	0
1069	200	0
441	2000	0
262	10000	0
214	20000	0
138	1.e+005	0
114	2.e+005	0
86.2	1.e+006	0

TABLE 39
Structural Steel > Strain-Life Parameters

Strength Coefficient MPa	Strength Exponent	Ductility Coefficient	Ductility Exponent	Cyclic Strength Coefficient MPa	Cyclic Strain Hardening Exponent
920	-0.106	0.213	-0.47	1000	0.2

TABLE 40
Structural Steel > Isotropic Elasticity

Temperature C	Young's Modulus MPa	Poisson's Ratio	Bulk Modulus MPa	Shear Modulus MPa
	2.e+005	0.3	1.6667e+005	76923

TABLE 41
Structural Steel > Isotropic Relative Permeability

Relative Permeability
10000